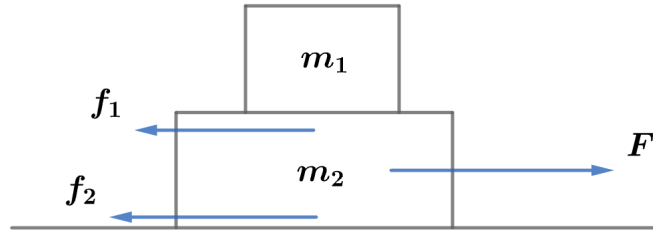


2010 F=ma Exam: Problem 10

Kevin S. Huang



Applying Newton's 2nd law to the lower block, we have

$$F - f_1 - f_2 = m_2 a_2$$

Recall kinetic friction is given by $f = \mu N$ so

$$f_1 = \mu m_1 g$$

$$f_2 = \mu(m_1 + m_2)g$$

Substituting into the first equation,

$$F - \mu(2m_1 + m_2)g = m_2 a_2$$

$$a_2 = \frac{F - \mu(2m_1 + m_2)g}{m_2}$$

so the answer is $\boxed{A}$.